

Project: ARAR HOSPITAL
Floor Name: Fourth Floor

Medical Gas Calculation

Oxygen	
HTM 02-01 Oxygen diversified flows (Table 13)	
Male Orthopedic Ward	
Nb of Terminal =	31 Terminals
Total Flow =	55 LPM
Female Orthopedic Ward	
Nb of Terminal =	28 Terminals
Total Flow =	50.5 LPM
In-patient Bed Ward	
Nb of Terminal =	28 Terminals
Total Flow =	50.5 LPM
Specialized Surgery Ward	
Nb of Terminal =	30 Terminals
Total Flow =	53.5 LPM

Vacuum	
HTM 02-01 Vacuum diversified flow (Table 21)	
Male Orthopedic Ward	
Nb of Bed Space=	31 Beds
Total Flow=	340 LPM
Female Orthopedic Ward	
Nb of Bed Space=	28 Beds
Total Flow=	310 LPM
In-patient Bed Ward	
Nb of Bed Space=	28 Beds
Total Flow=	310 LPM
Specialized Surgery Ward	
Nb of Bed Space=	30 Beds
Total Flow=	330 LPM

Summary 4th Floor

Total Oxygen Flow = 210 LPM
Total Vacuum Flow = 1290 LPM

Project: ARAR HOSPITAL
Floor Name: Thrid Floor

Medical Gas Calculation

Oxygen
HTM 02-01 Oxygen diversified flows (Table 13)
Male Orthopedic Ward
Nb of Terminal = 31 Terminals Total Flow = 55 LPM
Female Orthopedic Ward
Nb of Terminal = 28 Terminals Total Flow = 50.5 LPM
In-patient Bed Ward
Nb of Terminal = 28 Terminals Total Flow = 50.5 LPM
Specialized Surgery Ward
Nb of Terminal = 30 Terminals Total Flow = 53.5 LPM

Vacuum
HTM 02-01 Vacuum diversified flow (Table 21)
Male Orthopedic Ward
Nb of Bed Space= 31 Beds Total Flow= 340 LPM
Female Orthopedic Ward
Nb of Bed Space= 28 Beds Total Flow= 310 LPM
In-patient Bed Ward
Nb of Bed Space= 28 Beds Total Flow= 310 LPM
Specialized Surgery Ward
Nb of Bed Space= 30 Beds Total Flow= 330 LPM

Summary 3rd Floor

Total Oxygen Flow = 210 LPM
Total Vacuum Flow = 1290 LPM

Project: ARAR HOSPITAL
Floor Name: Second Floor

Medical Gas Calculation

Oxygen	
HTM 02-01 Oxygen diversified flows (Table 13)	
Medical ICU Department	
Nb of Bed Space=	6 Beds
Total Flow =	32.5 LPM
ICU Department	
Nb of Bed Space=	6 Beds
Total Flow =	32.5 LPM
Surgical ICU Department	
Nb of Terminal =	6 Beds
Total Flow =	32.5 LPM
CSICU Department	
Nb of Terminal =	6 Beds
Total Flow =	32.5 LPM
Burn Unit	
Nb of Terminal =	8 Beds
Total Flow =	41.5 LPM
Male DaySurgery	
Nb of OR Rooms =	25 Terminals
Total Flow =	46 LPM
Female DaySurgery	
Nb of Terminal =	25 Terminals
Total Flow =	46 LPM

Vacuum	
HTM 02-01 Vacuum diversified flow (Table 21)	
Medical ICU Department	
Nb of Bed Space=	6 Beds
Total Flow =	65 LPM
ICU Department	
Nb of Bed Space=	6 Beds
Total Flow =	65 LPM
Surgical ICU Department	
Nb of Terminal =	6 Beds
Total Flow =	65 LPM
CSICU Department	
Nb of Terminal =	6 Beds
Total Flow =	65 LPM
Burn Unit	
Nb of Terminal =	8 Beds
Total Flow =	75 LPM
Male DaySurgery	
Nb of OR Rooms =	25 Beds
Total Flow =	280 LPM
Female DaySurgery	
Nb of Terminal =	25 Beds
Total Flow =	280 LPM

Project: ARAR HOSPITAL
Floor Name: Second Floor

Medical Gas Calculation

Medical Air 400 kpa	
HTM 02-01 MA4 diversified flow (Table 18)	
Medical ICU Department	
Nb of Bed Space=	6 Beds
Total Flow =	280 LPM
ICU Department	
Nb of Bed Space=	6 Beds
Total Flow =	280 LPM
Surgical ICU Department	
Nb of Terminal =	6 Beds
Total Flow =	280 LPM
CSICU Department	
Nb of Terminal =	6 Beds
Total Flow =	280 LPM
Burn Unit	
Nb of Terminal =	8 Beds
Total Flow =	360 LPM

Summary 2nd Floor

Total Oxygen Flow = 263.5 LPM
Total Vacuum Flow = 895 LPM
Total MA4 Flow = 1480 LPM

Project: ARAR HOSPITAL
Floor Name: First Floor

Medical Gas Calculation

Oxygen	
HTM 02-01 Oxygen diversified flows (Table 13)	
Holding Bay	
Nb of Bed Space=	6 Terminals
Total Flow =	32.5 LPM
Operating Department	
Nb of Bed Space=	9 Rooms
Total Flow =	180 LPM
Recovery Bay	
Nb of Terminal =	14 Terminals
Total Flow =	88 LPM
Endoscopy Suite	
Nb of Terminal =	4 Terminals
Total Flow =	23.5 LPM
Endoscopy Suite (Recovery Bay)	
Nb of Terminal =	6 Terminals
Total Flow =	40 LPM

Vacuum	
HTM 02-01 Vacuum diversified flow (Table 21)	
Holding Bay	
Nb of Bed Space=	6 Terminals
Total Flow =	65 LPM
Operating Department	
Nb of Bed Space=	9 Rooms
Total Flow =	400 LPM
Recovery Bay	
Nb of Terminal =	14 Terminals
Total Flow =	170 LPM
Endoscopy Suite	
Nb of Terminal =	4 Terminals
Total Flow =	70 LPM
Endoscopy Suite (Recovery Bay)	
Nb of Terminal =	6 Terminals
Total Flow =	90 LPM

Medical Air 400 kpa	
HTM 02-01 MA4 diversified flow (Table 18)	
Operating Department	
Nb of Bed Space=	9 Rooms
Total Flow =	120 LPM
Recovery Bay	
Nb of Bed Space=	14 Beds
Total Flow =	72.5 LPM
Endoscopy Suite	
Nb of Terminal =	4 Terminals
Total Flow =	200 LPM
Endoscopy Suite (Recovery Bay)	
Nb of Bed Space=	6 Beds
Total Flow =	52.5 LPM

Surgical Air 700 kpa	
HTM 02-01 MA7 diversified flow (Table 20)	
Operating Department	
Nb of OR Rooms =	9 Rooms
Total Flow =	1750 LPM
CSSD	
Nb of Terminals=	2 Terminals
Total Flow =	42.5 LPM

Project: ARAR HOSPITAL
Floor Name: First Floor

Medical Gas Calculation

Nitrous Oxide	
HTM 02-01 N2O diversified flow (Table 15)	
Operating Department	
Nb of Bed Space=	9 Rooms
Total Flow =	63 LPM
Endoscopy Suite	
Nb of Bed Space=	3 Terminals
Total Flow =	13 LPM

Anasesthetic Gas Scavenging Sysyem	
HTM 02-01 AGSS diversified flow (Table 22)	
Operating Department	
Nb of Bed Space=	9 Rooms
Total Flow =	1170 LPM
Endoscopy Suite	
Nb of Bed Space=	3 Beds
Total Flow =	195 LPM

Summary 1st Floor

Total Oxygen Flow = 364 LPM
Total Vacuum Flow = 795 LPM
Total MA4 Flow = 445 LPM
Total MA7 Flow = 1792.5 LPM
Total N2O Flow = 76 LPM
TotalAGSS Flow = 1365 LPM

Project: ARAR HOSPITAL
Floor Name: Ground Floor

Medical Gas Calculation

Oxygen
HTM 02-01 Oxygen diversified flows (Table 13)
Exam/Triage Treatment Dept.
Nb of Terminal = 6 Terminals Total Flow = 17.5 LPM
Emergency (Ressuscitation)
Nb of Terminal = 8 Terminals Total Flow = 110.5 LPM
Emergency Room
Nb of Terminal = 2 Terminals Total Flow = 101.5 LPM
Emergency (Minor OR)
Nb of Terminal = 1 Terminals Total Flow = 10 LPM
Emergency (Recovery)
Nb of Terminal = 2 Terminals Total Flow = 14.5 LPM
Emergency (Observation)
Nb of Terminal = 10 Terminals Total Flow = 50.5 LPM
Emergency (ICU)
Nb of Terminal = 9 Beds Total Flow = 46 LPM
Emergency (Plaster)
Nb of Terminal = 1 Beds Total Flow = 10 LPM
Radiology Dept.
Nb of Terminal = 9 Terminals Total Flow = 26 LPM
Blood Donation.
Nb of Terminal = 3 Terminals Total Flow = 14 LPM

Vacuum
HTM 02-01 Vacuum diversified flow (Table 21)
Exam/Triage Treatment Dept.
Nb of Terminal = 6 Terminals Total Flow = 65 LPM
Emergency (Ressuscitation)
Nb of Terminal = 8 Terminals Total Flow = 110 LPM
Emergency Room
Nb of Terminal = 2 Terminals Total Flow = 50 LPM
Emergency (Minor OR)
Nb of Terminal = 1 Terminals Total Flow = 40 LPM
Emergency (Recovery)
Nb of Terminal = 2 Terminals Total Flow = 50 LPM
Emergency (Observation)
Nb of Terminal = 10 Terminals Total Flow = 85 LPM
Emergency (ICU)
Nb of Terminal = 9 Beds Total Flow = 80 LPM
Emergency (Plaster)
Nb of Terminal = 1 Beds Total Flow = 40 LPM
Radiology Dept.
Nb of Terminal = 9 Terminals Total Flow = 80 LPM
Blood Donation.
Nb of Terminal = 3 Terminals Total Flow = 50 LPM

Project: ARAR HOSPITAL
Floor Name: Ground Floor

Medical Gas Calculation

Medical Air 400 kpa	
HTM 02-01 MA4 diversified flow (Table 18)	
Emergency (Ressuscitation)	
Nb of Terminal =	8 Terminals
Total Flow =	75 LPM
Emergency Room	
Nb of Terminal =	2 Terminals
Total Flow =	45 LPM
Emergency (Minor OR)	
Nb of Terminal =	1 Terminals
Total Flow =	40 LPM
Emergency (Recovery)	
Nb of Terminal =	2 Terminals
Total Flow =	120 LPM
Emergency (ICU)	
Nb of Terminal =	9 Beds
Total Flow =	400 LPM
Emergency (Plaster)	
Nb of Terminal =	1 Beds
Total Flow =	80 LPM

Surgical Air 700 kpa	
HTM 02-01 MA7 diversified flow (Table 20)	
Emergency (Minor OR)	
Nb of OR Rooms =	1 Rooms
Total Flow =	350 LPM
Emergency (Plaster)	
Nb of Terminals=	1 Terminals
Total Flow =	40 LPM

Project: ARAR HOSPITAL
Floor Name: Ground Floor

Medical Gas Calculation

Nitrous Oxide	
HTM 02-01 N2O diversified flow (Table 15)	
Emergency (Minor OR)	
Nb of Bed Space=	1 Beds
Total Flow =	10 LPM
Emergency (Ressuscitation)	
Nb of Bed Space=	8 Beds
Total Flow =	20.5 LPM
Emergency (Recovery)	
Nb of Bed Space=	11 Beds
Total Flow =	25 LPM
Emergency (Plaster)	
Nb of Bed Space=	11 Beds
Total Flow =	25 LPM

Anasesthetic Gas Scavenging Sysyem	
HTM 02-01 AGSS diversified flow (Table 22)	
Emergency (Minor OR)	
Nb of Bed Space=	1 Beds
Total Flow =	130 LPM
Emergency (Ressuscitation)	
Nb of Bed Space=	8 Beds
Total Flow =	358 LPM
Emergency (Recovery)	
Nb of Bed Space=	11 Beds
Total Flow =	455 LPM
Emergency (Plaster)	
Nb of Bed Space=	11 Beds
Total Flow =	455 LPM

Summary Gr Floor

Total Oxygen Flow = 400.5 LPM
Total Vacuum Flow = 650 LPM
Total MA4 Flow = 760 LPM
Total MA7 Flow = 390 LPM
Total N2O Flow = 80.5 LPM
TotalAGSS Flow = 1397.5 LPM

Project: ARAR HOSPITAL

Medical Gas Calculation

Summary Hospital

	O2 Flow	Vac Flow	MA4 Flow	MA7 Flow	N2O Flow	AGSS Flow
	<i>LPM</i>	<i>LPM</i>	<i>LPM</i>	<i>LPM</i>	<i>LPM</i>	<i>LPM</i>
Fourth Floor	210	1290	0	0	0	0
Thrid Floor	210	1290	0	0	0	0
Second Floor	264	895	1480	0	0	0
First Floor	364	795	445	1793	76	1365
Ground Floor	401	650	760	390	81	1398
TOTAL	1447	4920	2685	2183	157	2763